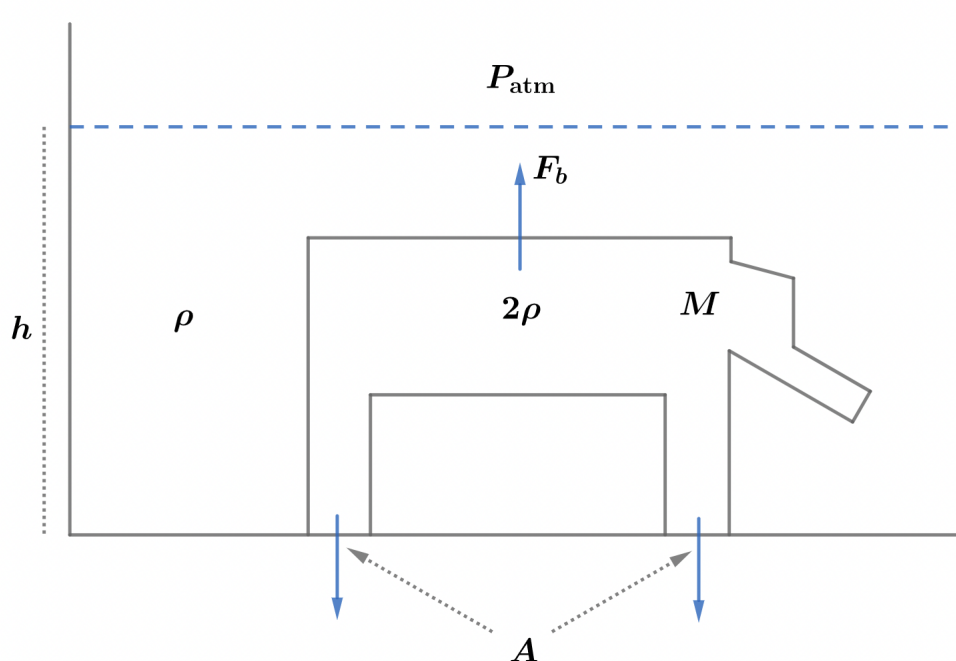


2022A F=ma Exam: Problem 19

Kevin S. Huang



If there was water under the elephant's legs, then the total hydrostatic force would be the buoyant force. Since the legs are on the floor, we subtract away the water pressure that would have been acting on the bottom of the legs:

$$F_{\text{hydro}} = F_b - PA$$

The buoyant force is

$$F_b = \rho V_{\text{elephant}} g = \rho \left(\frac{M}{2\rho} \right) g = \frac{Mg}{2}$$

The pressure at the bottom is

$$P = P_{\text{atm}} + \rho gh$$

Thus,

$$F_{\text{hydro}} = \frac{Mg}{2} - (P_{\text{atm}} + \rho gh)A = 0.6 \text{ N} - 1.6 \text{ N} = -1 \text{ N}$$

Since our sign convention takes upward to be positive, the answer is A.